

Horoscope and BaZi Memory Retrieval Agent

Abstract

This project implements the Horoscope and BaZi Memory Retrieval Agent, an AI agent designed to demonstrate how information retrieval improves contextual grounding in agent systems. The system combines a bilingual astrology and BaZi knowledge base, persistent user memory, temporal context retrieval, tool tracing, and an OpenAI-compatible model interface. The goal is not deterministic fortune telling, but a creative and inspectable agent architecture where retrieved context is visible before and after generation.

1 Introduction

The assignment asks for an AI agent that is extended in a useful way through information retrieval or tool use. This project uses horoscope and Chinese BaZi concepts as a creative domain for studying agent architecture. A normal chatbot may answer directly from the user prompt, while this agent first builds a context package from local knowledge, saved memory, and date-specific information.

The public repository is available at: <https://github.com/ousyu66-pixel/information-retrieval>. The video demonstration is available at: <https://drive.google.com/file/d/1sMcZSJjGHewY6A9Zf-VqF-zYQ5gEJ4tR/view?usp=sharing>.

2 System Design

The agent is implemented as a command-line Python application. It supports OpenAI-compatible chat completion APIs through three environment variables: `OPENAI_API_KEY`, `OPENAI_BASE_URL`, and `OPENAI_MODEL`. This allows the system to run with providers such as DeepSeek, while preserving the same interface expected by OpenAI-style clients.

The agent uses three main retrieval channels. First, it searches a local knowledge base containing Western astrology concepts such as zodiac signs, planets, houses, and moon phases, plus BaZi concepts such as BaZi, Five Elements, Heavenly Stems, Earthly Branches, Day Master, and Yong Shen. Second, it retrieves personal context from a persistent memory file, including profile data and previous forecast interactions. Third, it retrieves temporal context from structured date-based symbolic events.

The main tool loop is: extract or save profile information, read the user profile, search domain knowledge, retrieve relevant memory, retrieve temporal context, synthesize an answer with the model, and save the new forecast back to memory. Running the command with `-trace` exposes these tool calls, making the agent behavior visible for evaluation.

3 Information Retrieval Contribution

The main information retrieval contribution is the fusion of multiple context sources before generation. The domain retriever uses a lightweight BM25-style scoring method without external dependencies, which makes the retrieval logic easy to inspect. The memory retriever gives the system continuity across sessions, so later questions can use earlier interactions. The temporal retriever adds date-specific grounding, which is important for the horoscope use case.

The system is also designed to be extensible. New IR tools can be added as separate actions, for example document search, web search, vector retrieval, or a skill router. The BaZi extension makes the project more novel because it combines Western astrology and Chinese metaphysics as separate knowledge sources instead of treating the agent as a single-domain horoscope generator. The system also includes a small memory compaction mecha-

nism: when the event history grows, older information can be summarized while recent events remain directly searchable.

4 Video Interpretation

The video shows the agent running in a terminal with a DeepSeek model through an OpenAI-compatible API. The user asks for advice based on a birthday, study, relationships, BaZi, and Five Elements. The generated response is in Chinese, which demonstrates that the final synthesis is produced by the model API rather than only by the offline preview mode.

The trace demonstrates the actual agent loop. The tool `save_user_profile` extracts structured memory from natural language, including birth date, sun sign, and focus areas. The tool `get_user_profile` then reads that profile back into the context. The tool `search_astro_knowledge` retrieves relevant local knowledge documents such as Mercury symbolism, Taurus, Virgo, partnership themes, Yong Shen, and BaZi basics. These retrieved documents match the user's topics: study, relationship, personal astrology, and BaZi or Five Elements.

The tool `retrieve_user_memory` returns prior related runs from the memory file, showing continuity across sessions. The tool `get_temporal_context` retrieves date-specific information, especially the Mercury review window for the selected date. Finally, `save_forecast` stores the new answer back into memory. Therefore, the demo shows an agent using retrieval and tools, not just a direct chatbot response.

5 Reflection on AI Tool Use

I used AI coding tools to help build the overall framework of this project. During the process, I gained a clearer understanding of how an AI agent works in practice: it receives a user request, extracts useful state, calls tools, retrieves context, sends the organized context to a model, and saves useful information back to memory. At the same time, I am aware that there are still parts of the code-level implementation that I do not fully understand yet, especially the details of retrieval scoring, memory management, and API integration. This is something I need to continue studying carefully, because using AI to generate code is not enough; I also need to understand, check, and maintain the system myself.

6 Conclusion

This project helped me understand an AI agent. The agent built in this project receives a user question, extracts useful personal context, retrieves relevant knowledge, checks saved memory, adds date-specific temporal information, and then uses an OpenAI-compatible model API to produce the final response. Through this process, the project shows how information retrieval can make an agent's answer more grounded, explainable, and reusable across sessions.

The project also shows that tool use is an important part of agent design. The visible trace makes it possible to see which tools were called and which retrieved documents influenced the final answer. By combining Western astrology, BaZi concepts, memory, and temporal context, I learned how different sources of information can be organized before generation instead of being hidden inside the model. A future version could improve the system with larger knowledge sources, better retrieval scoring, and a clearer user interface.

References

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